

Cutaneous melanoma in childhood and adolescence shows frequent loss of INK4A and gain of KIT.

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Childhood cutaneous melanoma is a rare disease with increasing incidence. It is not clear whether it differs from adult melanoma in etiology and clinical evolution. To genetically characterize childhood melanoma, 21 pediatric patients were studied by germ-line analysis of CDKN2A, CDK4, and MC1R genes. In addition, alterations in CDKN2A, c-Kit, BRAF, and NRAS genes were evaluated at the somatic level by direct gene sequencing, fluorescence in situ hybridization analysis, and immunohistochemistry. As a control group of susceptible patients, we studied patients from 23 melanoma-prone families. At the germ-line level, CDKN2A and MC1R gene variants were detected in 2/21 and 12/21 pediatric patients and in 9/23 and 19/22 in familial patients. At the somatic level, most lesions (9/14) from pediatric patients showed CDKN2A locus homozygous deletions and a null p16 immunophenotype, whereas most lesions (5/8) from familial patients were disomic and immunoreactive. A c-Kit low-polysomy profile seems to parallel CDKN2A homozygous deletions in pediatric melanoma whereas the single activating mutation observed segregates with familial patients. Loss of KIT protein expression was frequent (7/14) in pediatric melanomas, where metastatic cases were prevalent. BRAF(V600E) mutation occurred at a similar rate (approximately 50%) in lesions from pediatric and familial patients, whereas no NRAS mutations were detected.

Exon 15 BRAF mutations are uncommon in canine oral malignant melanomas.

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An activating mutation in codon 599 of BRAF has been identified in approximately 60% of human cutaneous nevi and melanomas, but not melanomas of mucosal origin. The purpose of this study was to determine if BRAF mutations occur in canine oral malignant melanomas. The canine BRAF gene was first cloned from normal canine testicular cDNA, and a novel previously unreported splice variant involving exon 5 was identified during this process. To screen canine melanoma samples for BRAF mutation in codon 599, cDNA and genomic DNA were isolated from canine malignant melanoma cell lines and primary tumor samples respectively, all from cases seen at the Veterinary Medical Teaching Hospital at the University of California, Davis. Polymerase chain reaction (PCR) was performed for exon 15 using primers based at the 5' end of exon 15 and the 5' end of intron 15 and the resultant products were directly sequenced. No mutations in codon 599 or exon 15 were identified in any of the 17 samples evaluated. However, all of the melanoma cell lines expressed BRAF and demonstrated high levels of basal ERK phosphorylation suggesting that dysregulation of this pathway is present. Therefore, similar to the case with human mucosal melanomas, canine oral malignant melanomas do not possess codon 599 BRAF mutations commonly identified in human cutaneous melanomas. This finding supports the notion that melanomas arising from non-sun-exposed sites exhibit distinct mechanisms of molecular transformation.

BRAF inhibitor discontinuation and rechallenge in advanced melanoma patients with a complete initial treatment response.

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BRAF inhibitors (BRAFi), a targeted therapy, are used to treat metastatic late-stage melanomas harbouring the BRAF-V600 mutation (found in about 50% of melanomas). The targeted therapy is generally maintained until tumour progression or major toxicity occurs, although responses are often limited in time. It is unknown whether melanoma patients achieving a complete response with targeted therapy can safely discontinue treatment. We retrospectively observed the clinical course of patients with metastatic melanoma who discontinued BRAFi therapy after achieving a complete response and those with an incomplete response combined with surgical removal of the remaining tumours. We also evaluated the effectiveness of BRAFi in these patients after recurrence. In 11 patients, the best response was diagnosed after a median BRAFi treatment duration of 105 (29-341) days. The median follow-up after BRAFi initiation was 769 (435-1765) days. Recurrence was observed in all 11 patients (100%), median: 82 (27-322) days. Five patients achieved a complete response, with a median progression-free survival after cessation of 136.5 (34-322) days versus 82 (27-144) days for six patients with an incomplete response combined with surgical removal of remaining tumours. Baseline characteristics and time to best response and to discontinuation did not influence the rate of relapse. Subsequently, eight patients were rechallenged with a BRAFi. The median progression-free survival time after BRAFi rechallenge was 222.5 (15-425) days. The three remaining patients received treatments other than BRAFi. Our findings may be valuable with respect to ongoing clinical trials of combinations of targeted therapies and immunomodulatory antibodies.

There is a world beyond protein mutations: the role of non-coding RNAs in melanomagenesis.

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Until recently, the general perception has been that mutations in protein-coding genes are responsible for tumorigenesis. With the discovery of (V600E)BRAF in about 50% of cutaneous melanomas, there was an increased effort to find additional mutations. However, mutations characterized in melanoma to date cannot account for the development of all melanomas. With the discovery of microRNAs as important players in melanomagenesis, protein mutations are no longer considered the sole drivers of tumors. Recent research findings have expanded the view for tumor initiation and progression to additional non-coding RNAs. The data suggest that tumorigenesis is likely an interplay between mutated proteins and deregulation of non-coding RNAs in the cell with an additional role of the tumor environment. With the exception of microRNAs, our knowledge of the role of non-coding RNAs in melanoma is in its infancy. Using few examples, we will summarize some of the roles of non-coding RNAs in tumorigenesis. Thus, there is a whole world beyond protein-coding sequences and microRNAs, which can cause melanoma.

Managing Side Effects of Vemurafenib Therapy for Advanced Melanoma.

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Somatic point mutations in the BRAF gene have been found in approximately 50% of melanomas. BRAF (V600E), the most common mutation, results in the constitutive activation of BRAF (V600E) kinase, sustaining MAPK signaling and perpetuating cell growth. This groundbreaking discovery led to the clinical development of vemurafenib, a selective BRAF inhibitor. Vemurafenib has been approved for the

treatment of patients with BRAF (V600E)-positive unresectable or metastatic melanoma based on survival benefit demonstrated in a randomized phase III study. The current approved dosing schedule of vemurafenib is 960 mg orally twice a day until disease progression or unacceptable toxicity. Vemurafenib is well tolerated, with the most common adverse effects including skin reactions, photosensitivity, headache, and arthralgia. Active research is ongoing to expand the utility of vemurafenib into the adjuvant setting and to circumvent rapid emergence of drug resistance.

Dabrafenib in advanced melanoma with BRAF V600E mutation.

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Dabrafenib was recently approved by the US Food and Drug Administration for treatment of unresectable or metastatic melanoma with BRAF V600E mutations as detected by an FDA-approved test. The THxID BRAF assay, for detection of BRAF V600E mutations was concurrently approved. Dabrafenib is not indicated for the treatment of patients with wild-type BRAF melanoma, because of the potential risk of tumor promotion. About 50% of melanomas have an activating mutation in the BRAF gene, with about 80%-90% of those having a V600E mutation, and 10%-20% having a V600K mutation. Dabrafenib is a reversible, ATP-competitive inhibitor that selectively inhibits BRAF V600E kinase; preclinical data indicate that dabrafenib inhibits the MAPK pathway in BRAF V600E-mutated melanoma cells, leading to decreased proliferation and regression in xenograft models. Dabrafenib also inhibits other mutated forms of BRAF kinases, including BRAF V600K and BRAF V600D enzymes and, at higher concentrations, wild-type BRAF and CRAF kinases and other kinases (eg, SIK1, NEK11, and LIMK1). However, in vitro experiments have shown paradoxical activation of MAP-kinase signaling and increased cell proliferation in BRAF wild-type cells exposed to BRAF inhibitors.

Multifocal BRAF(V600E)-Mutated Melanoma in situ on the Foot.

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Melanoma is an aggressive skin cancer that originates from melanocytes, and about one half of melanoma cases possess a BRAF mutation. Together with PD-L1 expression, the BRAF(V600E) mutation is one of the optimal therapeutic targets for the treatment of melanoma. In this report, we describe a case of multifocal melanoma in situ on the foot, which carried the p.V600E mutation in the BRAF gene. Interestingly, the spotted melanoma lesion is demarcated by normal skin, and in all spotted pigmented lesions, there were no signs of dermal invasion of melanoma cells or spontaneous regression. Our case presented atypical clinical features, which might correlate with the local mutations of BRAF gene and the immunological expression of PD-L1.

The \textit{BRAF} V600E mutation in single- institution study of Russian melanoma patients.

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The activating mutation BRAF V600E is considered to be a diagnostic cutaneous melanoma (CM) marker

important for prognosis and targeted therapy. The aim of this study was to determine the frequency of the V600E mutation in CM patients in Russia and to estimate the influence of the BRAF gene mutation status on prognosis and clinical outcome. To ensure mutation detection in FFPE tissue, interlaboratory validation was performed using three different methods: allele-specific hybridisation on a biochip, allele-specific real-time PCR and, in some cases, direct sequencing. Mutation V600E was detected in 49% of patients. The age of disease manifestation was significantly lower in mutated (MT) BRAF patients, and the median age difference between the wild-type (WT) and MT BRAF groups ($P = 0.002$) was 10 years. A tumour thickness more than 1 mm was also more frequently observed in the MT BRAF group ($P = 0.059$). Patients from the MT BRAF group were more likely to have ulceration compared to the WT group ($P = 0.088$). No statistically significant differences were found between the relapse-free, progression-free or overall survival of CM patients in the MT BRAF and WT BRAF groups. The data obtained show that the V600E BRAF mutation occurred in about half of melanoma patients; it was associated with earlier manifestation of melanoma and likely with more aggressive clinical features.

Sensitive detection of BRAF V600E mutation by Amplification Refractory Mutation System (ARMS)-PCR.

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BRAF mutations occur in approximately 8% of all human cancers and approach 50% in melanoma and papillary carcinoma of thyroid. These mutations provide potentially valuable diagnostic, prognostic and treatment response prediction markers. A sensitive, specific, low-cost assay to detect these mutations is needed. To detect BRAF V600E mutation in formalin-fixed, paraffin-embedded (FFPE) tissue, we developed a method using Amplification Refractory Mutation System (ARMS)-PCR. This method was designed to amplify three products in a single reaction tube: a 200 bp common product serving as an amplification control, a 144 bp BRAF V600E specific product, and a 97 bp wild-type (wt) specific product. The sensitivity of this method was determined to be as low as 0.5% for the BRAF V600E allele in a wild-type background. This method was successfully validated in 72 thyroid tumors. It detected V600E mutation in 22 out of 33 (67%) of the conventional papillary thyroid carcinoma (PTC), 8 out of 12 (75%) of the tall-cell variant of PTC, whereas none of the 10 follicular variant of PTC showed BRAF V600E mutation. In addition, none of the 14 follicular adenomas and 3 follicular carcinomas had BRAF V600E mutation. As a comparison method, direct dideoxy sequencing found only 27 out of 30 (90%) mutations detected by ARMS-PCR method, suggesting that this ARMS-PCR method has higher sensitivity. Our ARMS-PCR method provides a new tool for rapid detection of BRAF V600E mutation. Our results indicate that ARMS-PCR is more sensitive than automated dideoxy sequencing in detecting low BRAF V600E allele burdens in FFPE tumor specimen. The strategy of this ARMS-PCR design may be adapted for early detection of point mutations of a variety of biomarker genes.

Frequency of BRAF V600E Mutation in the Mexican Population of Patients With Metastatic Melanoma.

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The BRAF V600E mutation has been described in melanomas occurring in the Caucasian, European, and Asian populations. However, in the Mexican population, the status and clinical significance of BRAF

mutation has not been researched on a large scale. Consecutive BRAF-tested Mexican patients with metastatic melanoma (n = 127) were analyzed for mutations in exon 15 of the BRAF gene in genomic DNA by real-time polymerase chain reaction technology for amplification and detection. The results were correlated with the clinical-pathologic features and the prognosis of the patients. The frequency of somatic mutation V600E within the BRAF gene was 54.6% (43 of 127 patients). Nodular melanoma was the most prevalent subtype in our population, with BRAF mutations in 37.2% (16 of 55 patients). In contrast, superficial spread had a frequency of 18.6% BRAF mutation (eight of 24). Other clinicopathologic features were assessed to correlate with the mutation status. This study searched for the most prevalent BRAF V600E mutation type in melanoma in a heterogeneous population from Mexico. Nodular melanoma was found to be the most prevalent in metastatic presentation and the presence of BRAF V600E mutation, perhaps related to the mixed ancestry; in the north, ancestry is predominantly European and in the south, it is predominantly Asian. The outcomes of the mutation correlations were similar to those found in other populations.
